Wh-conditionals are more interrogative than not

An insight from Khorchin Mongolian

[疑問代詞條件句的疑問特徵 — 來自科爾沁蒙古語的启發]

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Abstract The observation that *wh*-expressions in many languages can be interpreted either interrogatively or non-interrogatively has led to the widely accepted thesis that *wh*-expressions originate as variables, with their interpretations determined through being bound by a question operator or other non-interrogative operators (Cheng 1991; Li 1992; Tsai 1994; Cheng & Huang 1996; among others). However, recent studies on a specific non-interrogative *wh*-construction in Mandarin—*wh*-conditionals—challenge this variable-binding approach (Liu 2016, 2017; Xiang 2016, 2020; Li 2021). In *wh*-conditionals, two *wh*-clauses are connected to express a conditional meaning. These studies advocate for a question-based approach, proposing that the meaning of *wh*-conditionals stems from interrogative meaning rather than variable binding. As a result, the question-based approach predicts that *wh*-conditionals should be linguistically closer to *wh*-questions than to other non-interrogative *wh*-constructions. This paper confirms the prediction using *wh*-constructions in Khorchin Mongolian. Despite their distinct theoretical commitments, the two approaches can be reconciled to address challenges pointed out by Cheng & Huang (2020), Tsai (2023), and Pan & Kuang (2023) to the question-based view.

Keyword *wh*-conditionals · *wh*-indefinites · the variable binding approach · the question-based approach · alternative semantics · Khorchin Mongolian

關鍵詞: 疑問代詞條件句、疑問代詞的無定用法、變量約束分析、疑問句基底分析、選項語義學、科爾沁蒙古語

1. Introduction

Cross-linguistically, *wh*-expressions are used as interrogative items or as non-interrogative indefinites. Haspelmath (1997) shows that *wh*-indefinites occur in 64 languages out of 100 languages in his sample. Mandarin provides clear examples of this phenomenon:

(1) Libai diu-le shenme dongxi?
Libai lose-PFV what thing
'What did Libai lose?' Interrogative

(2) Libai haoxiang diu-le shenme dongxi.
Libai likely lose-PFV what thing
'It seems that Libai lost something' Non-interrogative

The prevailing assumption about this phenomenon is that interrogative and non-interrogative *wh*-expressions share a common 'indeterminate' denotation and the various interpretations of *wh*-constructions are 'determined' by other operators at the clausal level. A traditional approach holds that *wh*-expressions are interpreted as variables—either individual variables or choice function variables—that are bound by various operators (Cheng 1991; Li 1992; Tsai 1994, 1999; Cheng & Huang 1996; Lin 1998, 1999, 2004; Slade 2011, among others).

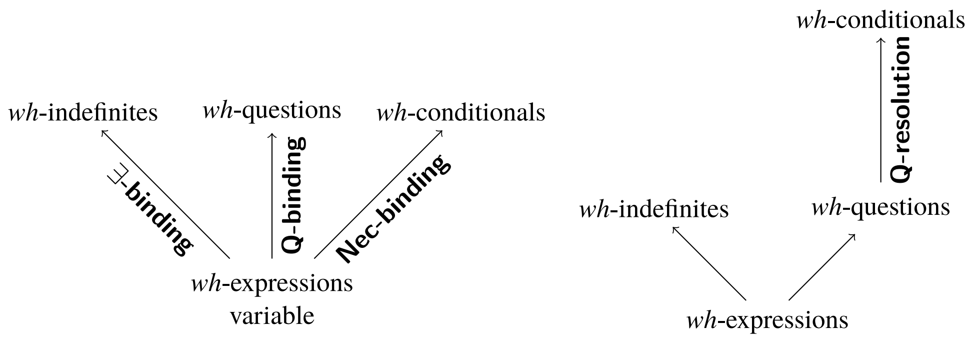
While the homogeneous view is insightful and has shed light on the cross-linguistic affinity between interrogatives and indefinites, recent research on a specific non-interrogative *wh*-construction—namely, *wh*-conditionals, sometimes referred to as 'Chinese donkey sentences' following Cheng & Huang (1996)—has pointed toward a **heterogeneous** perspective. Consider the Mandarin example in (3), where two *wh*-clauses combine to form a conditional. The adverb *jiu* 'then' is typically analyzed as a marker of conditionality in Chinese linguistics.

(3) Shei shu-le, shei jiu qingke.
who lose-PFV who then treat
'For everyone, if they lose the bet, they will treat us.'

The classical approach to *wh*-conditional, which is initiated by Cheng & Huang (1996) and widely adopted (Tsai 1994, 1999; Lin 1996, 1999; Chierchia 2000; Pan & Jiang 2015; Cheng & Huang 2020; among others), treats the *wh*-expressions in (3) similarly to that in (2)—they are all analyzed as variables. Accordingly, the *wh*-expressions in (3) are unselectively bound by the necessity modal in the conditional, resulting in a universal statement: for all individuals x and all possible worlds w, if x loses the bet in w, then x treats us in w. In contrast, a more recent question-based approach, advocated by Liu (2016, 2017), Xiang (2016, 2020), and Li (2020, 2021), suggests that the *wh*-conditionals express resolution dependencies between *wh*-questions. For example, (3) involves two *wh*-questions and conveys that for all individuals x and possible worlds w, if x resolves the issue raised by the question *who loses* in w, then x resolves the issue raised by the question *who treats everyone* in w. Under this view, *wh*-conditionals are built up on the basis of *wh*-questions and do not share a common orign with *wh*-indefinites.[1]

The two approaches can be illustrated by the diagrams in Figure 1, each leading to distinct predictions about the relations among *wh*-constructions. The classical analysis, rooted in the variable-binding approach, predicts that *wh*-conditionals are fundamentally non-interrogative, as the involved *wh*-expressions are not bound by the question operator, similar to *wh*-indefinites. As a consequence, *wh*-conditionals should not exhibit hallmarks of *wh*-questions. Instead, the question-based approach posits that *wh*-conditionals are derived from *wh*-questions and

[1] In the literature, there is also a 'correlative/free-relative' approach to wh-conditionals, which interprets wh-conditionals as *ever*-free relatives (Bruening & Tran 2006; Luo & Crain 2011; Huang 2010). Since this approach offers little insight into *wh*-indefinites, it will not be reviewed in this paper. However, structural and interpretive issues with this approach have been noted by Cheng & Huang (1996) and Liu (2016).



(a) Variable binding approach **(b)** Question-based approach (to be revised)

Figure 1: Two approaches to unifying *wh*-constructions

are independent of *wh*-indefinites. Consequently, this approach gains strong support if *wh*-conditionals are demonstrated to be linguistically closer to *wh*-questions than to *wh*-indefinites.

The homogeneous view, grounded in the variable-binding approach, is largely motivated by the fact that in Mandarin, *wh*-expressions in *wh*-conditionals do not display any systematic differences from their indefinite counterparts. There are three notable exceptions: the degree *wh*-expression *duo(shao)* 'how many', the reason *wh*-expression *weishenme* 'why', and manner *wh*-expression *zenme* 'how' (Tsai 1994; Lin 1996; Li 2021). They can appear in both *wh*-questions and *wh*-conditionals, but cannot be used as *wh*-indefinites when embedded in the antecedent clause of a conditional, as illustrated below.[2]

(4) a. *Ruguo Libai weishenme bei jiegu, ta yiding hui lai gaosu wo.
if Libai why PASS fire he must will come tell me

[2] One of the reviewers notice that *zenme* and *duoshao* may be interpreted as indefinites in other structural environments, as shown below.

(i) Libai mei he duoshao jiu.
Libai not drink how.much wine
'Libai didn't drink too much wine.'

(ii) Ni keyi suibian chi duoshao dongxi.
you can whatever eat how.many thing
'You can eat as much as you want.'

(iii) Zhe-dao cai hen jiandan, ni keyi suibian zenme zuo.
this-CL dish very simple you can whatever how do
'This dish is simple and you can make it any way you want.'

Indeed, *duoshao* can receive an indefinite interpretation under the scope of negation. However, other typical licensors of indefinites—such as conditionals, polar questions, and modals—do not license its indefinite use. In (ii) and (iii), the modal *keyi* 'can' does not appear to serve as the licensor; rather, the adverb *suibian* 'whatever' plays that role. Once *suibian* is removed, these sentences lose their non-interrogative, existential readings. Notably, inserting *suibian* into the examples in (4) fails to rescue them. While a full account of the *wh*-indefinites in (i) through (iii) lies beyond the scope of this paper, these cases highlight the need for a more consistent contrast between *wh*-indefinites and *wh*-conditionals to strengthen his argument.

Intended: 'If Libai was fired for some reason, he surely would come to tell me.'

b. *Ruguo Libai zenme xiuhao-le che, ta yiding hui hen gaoxing.
if Libai how fix-PFV car he must will very happy
Intended: 'If Libai fixed the car in some way, he must be very happy.'

c. *Ruguo Libai diu-le duoshao dongxi, ta yiding hui lai gaosu wo.
if Libai lose-PFV how.many thing he must will come tell me
Intended: 'If Libai lost some number of items, he surely would come to tell me.'

(5) a. Libai weishenme bei jiegu, Dufu jiu weishenme bei jiegu.
Libai why PASS fire Dufu then why PASS fire
'For every x, if x is the reason why L. was fired, x is the reason why D. was fired.'

b. Libai zenme xiu che, Dufu jiu zenme xiu che.
Libai how fix car Dufu then how fix car
'For every x, if x is the way that L. fixes cars, x is the way that D. fixes cars.'

c. Libai chi-le duoshao, Dufu jiu chi-le duoshao.
Libai eat-PFV how.many Dufu then eat-PFV how.many
'For every x, if x is the amount of food L. ate, x is the amount of food D. ate.'

These exceptions may be attributable to independent factors that constrain the non-interrogative uses of certain *wh*-expressions. As one reviewer suggests, under the exhaustification approach (Liao 2011; Chierchia & Liao 2015; Law 2018; Liu & Yang 2021, among others), the indefinite interpretation of a *wh*-expression depends on the set of alternative domains it evokes—for example, an instance of *who* ranging over a domain D gives rise to alternatives ranging over proper subdomains D' of D. The absence of indefinite readings in (4) may therefore be due to the failure of these specific *wh*-expressions to activate such alternative domains.

Although in Mandarin *wh*-indefinites are not distinct from interrogative *wh*-expressions regarding form, this is not necessarily the case in other languages. As noted by Haspelmath (1997), many languages, like Hebrew, Romanian, Basque, and Japanese, employ unmarked forms for interrogative *wh*-expressions, while *wh*-indefinites require additional marking. If the question-based approach is correct, we would expect *wh*-expressions in *wh*-conditionals to pattern with interrogative *wh*-expressions rather than with *wh*-indefinites regarding form.

While *wh*-indefinites are widely attested across languages, *wh*-conditionals are rarely documented. As a result, existing arguments for both the variable-binding and question-based approaches have been developed primarily with reference to Mandarin, leaving a cross-linguistic perspective largely unexplored. Fortunately, a comparative study of Tungusic languages by Baek (2016) identifies the presence of *wh*-conditionals in Mongolian, providing an opportunity to evaluate these two approaches using novel data. In this paper, we investigate *wh*-constructions in Khorchin Mongolian—a variety of Mongolian spoken in the east of Inner Mongolia—and show that the language exhibits the following properties.

- *Wh*-questions, *wh*-indefinites, and *wh*-conditionals are all allowed;
- *Wh*-indefinites differ from *wh*-expressions in *wh*-questions and *wh*-conditionals: the former must be followed by a dedicated morpheme, whereas the latter systematically exclude it;

- *Wh*-expressions in *wh*-conditionals display the same structural and interpretive properties as those in *wh*-questions.

Based on these facts, we argue for the question-based approach—in Khorchin Mongolian, *wh*-conditionals are formed from two *wh*-clauses with interrogative denotations, whereas *wh*-indefinites are interpreted as non-interrogative existential quantifiers.

While this paper advocates for the question-based approach, it also highlights its connection to the variable-binding approach, suggesting that the former can be viewed as an extension of the latter—one that accounts for a broader range of empirical patterns while preserving its core insights. As a result, certain challenges to the question-based approach are naturally addressed.

This paper is structured as follows. Section 2. provides a brief overview of Mongolian grammar to lay the foundation for the analysis. Section 3., which forms the core of the paper, compares *wh*-questions, *wh*-indefinites, and *wh*-conditionals in Khorchin Mongolian. Section 4. presents a semi-formal analysis that reflects the spirit of the question-based approach while illustrating its connection to the variable-binding approach. Finally, Section 5. summarizes the main findings and concludes the paper.

2. A brief introduction to Mongolian

Mongolian is the principal language of Mongolic languages, as a member of the Altaic family. The majority of its speakers live in Mongolia, China, and Russia. Mongolian contains a number of dialects, the classification of which has been controversial, with no universal approval available. Janhunen (2012) proposes that the Mongolic language family consists of six branches, among which the Common Mongolian is divided into six dialects including Khalkha, spoken primarily in Mongolia, Khorchin, spoken in eastern Inner Mongolia and its adjacent areas, Ordos, spoken in Ordos City in Inner Mongolia, Khamnigan, spoken in an area divided between Russia (Transbaikalia), China (Hulun Buir) and Mongolia (Khentei Aimak), Buryat, spoken on both sides of Lake Baikal in the Republic of Buryatia of Russia, and Oirat, spoken by people living in the west of Dzungaria Banner of Inner Mongolia. However, Buryat, Oirat, and Khammigan are sometimes considered as independent languages. The data in this paper were collected from **Khorchin Mongolian**, but the target phenomena—*wh*-constructions—do not exhibit notable variation at least across Khorchin and Khalkha Mongolian languages, whose speakers comprise a large proportion of the whole family of speakers.

Mongolian is generally an agglutinative, head-final language in which verbs and their verbal suffixes consistently appear to the right of the phrases or clauses they head. These verbal suffixes encode tense, aspect, and voice. Additionally, the nominal system in Mongolian features case inflections. Specifically, six distinct cases are marked by dedicated suffixes: genitive (*-in*), dative (*-d*), accusative (*-ig*), ablative (*-aas*), instrumental (*-aar*), and comitative (*-tai*), while the nominative case remains unmarked. The following examples demonstrate some of these basic features.[3]

[3] Phonemic transcriptions of Mongolian alphabets lack standardized regulation, often leading to different letters being

(6) Bi Bat-ig ger-t-ni hari-uul-čih-san.
ISG.NOM Bat-ACC home-DAT-POSS return-CAUS-PFV-PST
'I have let Bat go home.'

(7) Bat-in düü Bat-tai hamt Shanghai-aas mašin-aar ir-sen.
Bat-GEN.NOM brother Bat-COM together Shanghai-ABL car-INS come-PST
'Bat's brother came from Shanghai by car with Bat.'

A particularly notable feature of Mongolian grammar is the case marking of embedded subjects, which differs from that of matrix subjects. Subjects of complement clauses may be marked with either genitive or accusative case, primarily determined by the lexico-syntactic properties and (non)finiteness of the matrix verbs. Nominative marking is also possible but less common. In relative clauses, overt subjects are consistently genitive-marked. For adjunct clauses, subjects are predominantly marked with accusative case; nominative marking is less common, and genitive marking is not permitted.

(8) Bi Bat-ig ir-ne gež bod-san güi.
1SG.NOM Bat-ACC come-NPST COMP think-PST not
'I did not think that Bat would come.'

(9) Bi Bat-in ög-sen nom-ig gee-čih-sen.
1SG.NOM Bat-GEN give-PST book-ACC lose-PFV-PST
'I lost the book that Bat gave me.'

(10) Bi Bat-ig ir-h-d negent yab-čih-san.
1SG.NOM Bat-ACC come-COV-DAT already go-PFV-PST
'I had left when Bat came.'

3. *Wh*-constructions in Khorchin Mongolian

This section discusses *wh*-questions, *wh*-indefinites, and *wh*-conditionals in Khorchin Mongolian, highlighting that *wh*-expressions in *wh*-conditionals closely parallel interrogative *wh*-

used to represent the same vowel or consonant. In this paper, we adopt the following conventions: (a) 'o', 'u', 'ö', and 'ü' represent the fourth, fifth, sixth, and seventh vowels, respectively; (b) č, š, and ž represent the ninth, twelfth, and thirteenth consonants, for which some authors have alternatively used 'ch', 'sh', and 'j'. Moreover, in presenting Mongolian data, we adopt the convention widely used in modern linguistics: word stems are consistently separated from grammatical suffixes, with a hyphen marking the stem – suffix boundary. This practice departs from the traditional Mongolian script, in which nominal stems and inflectional suffixes are typically written separately (e.g., case suffixes), but in certain cases are written contiguously (e.g., the plural suffix *-d*), as shown below. The uniform use of hyphenation, however, makes the morphological structure of Mongolian more transparent.

Traditional Mongolian script	Latin transcription of traditional Monoglian script	Cyrillic script	This paper	Gloss
ᠮᠣᠷᠢ ᠶᠢ	mori gi	морийг	mori-ig	horse-ACC
ᠮᠣᠷᠢᠳ	morid	морьд	mori-d	horses

In addition, the accusative marker in Mongolian may be written as *-ig* or *-iig*. For consistency, this paper uniformly adopts *-ig*.

expressions in many respects, while diverging from *wh*-indefinites.

3.1. *Wh*-questions

Typologically, Mongolian is a *wh*-in-situ language, as demonstrated by the following examples. Note that the question particle *be* is required to mark matrix *wh*-questions.

(11) Bat **yuu** id-sen be?
Bat.NOM what eat-PST Q
'What did Bat eat?'

(12) Önödör **hen** ir-sen güi be?
Today who.NOM come-PST not Q
'Who didn't come today?'

(13) Bat **ali** tom od-ig har-san be?
Bat.NOM which big pop.star-ACC see-PST Q
'Which pop star did Bat see?'

(14) Bat **hežee** Beejing-d oči-h be?
Bat.NOM when Beijing-DAT go-INF Q
'When will Bat go to Beijing?'

(15) Bat ene dugui-g **yaaž** zas-san be?
Bat.NOM this bike-ACC how repair-PST Q
'How did Bat repair this bike?'

(16) Bat **yuund/yaagad** ene üy-d [detegber-d gar]-san be?
Bat.NOM for.what/why this time-DAT retire-PST Q
'Why does Bat retire now?'

The question particle *be* also appears in embedded questions if embedded questions are selected by the so-called 'complementizer' *gĕz*, as illustrated in (20); whereas, once *gež* is omitted, *be* cannot appear, and *wh*-questions may be constructed in a different way, as shown in (18). For these examples, while the two sentences differ in syntax, they convey roughly the same meaning.

(17) Bat [Dorž-oos **yuu** id-sen be gež] asuu-san.
Bat.NOM Dorž-ABL what eat-PST Q COMP ask-PST
'Bat asked Dorž what he ate.'

(18) Bat [Dorž-in **yuu** id-sen]-ig asuu-san.
Bat.NOM Dorž-GEN what eat-PST-ACC ask-PST
'Bat asked what Dorž ate.'

Unsurprisingly, a *wh*-question can also be embedded under a responsive verb like *med* 'know', as observed in (19). Unlike rogative verbs, however, responsive verbs do not permit their complement questions to contain the question particle.[4]

[4] A reviewer points out that question-embedding verbs like *know* can take either an interrogative clause or a definite description. This raises a potential confound: the embedded *wh*-constituent in (19) might not be interrogative but

(19) Bat [Dorž-in **yuu** id-sen-ig] med-ne.
Bat.NOM Dorž-GEN what eat-PST-ACC know-NPST
'Bat knows what Dorž ate.'

(20) *Bat [Dorž-ig **yuu** id-sen be gež] med-ne.
Bat.NOM Dorž-ACC what eat-PST Q COMP know-NPST
Intended 'Bat knows what Dorž ate.'

To our knowledge, there has not been a comprehensive investigation into the differences between these two types of embedded questions.[5] However, the aspect relevant to the present study is that a *wh*-clause can be interpreted interrogatively in the absence of a question particle. This can be analogous to the distribution of *do*-support in English *wh*-questions: while *do*-support is required in matrix *wh*-questions, it is prohibited in embedded contexts. Crucially, the absence of *do*-support in embedded *wh*-questions does not undermine their interrogative status.

3.2. *Wh*-indefinites

In Khorchin Mongolian, *wh*-expressions can be used as non-interrogative indefinites only when they co-occur with the numeral item *neg(en)* 'one', as shown in (21).[6] If *neg(en)* is omitted, the sentence can only be interpreted as a *wh*-question, as in (22).

(21) Bat **hen negen**-ig uurluul-san.
Bat-NOM who one-ACC irritate-PST
'Bat irritated someone.'

(22) Bat **hen** uurluul-san be.
Bat-NOM who irritate-PST Q
'Bat irritated someone.'

It shows that Mongolian *wh*-indefinites can occur without the presence of licensing elements such as modals, negation, or polar question particles. (21) straightforwardly asserts the existence of a person who Bat irritated. This patterns with German *wh*-indefinites (Haida 2007; Hengeveld et al. 2023), but differs from the widely accepted generalization about Mandarin *wh*-indefinites, which are typically analyzed as polarity items and are excluded from standard

rather a nominal expression referring to the food Dorž ate. While this concern is reasonable, there are fair reasons to reject the view that a *wh*-constituent without an overt question particle is simply a non-interrogative nominal phrase. First, like Mandarin, which distinguishes between acquaintance and attitude verb uses of *know* (e.g., *rènshí*), Mongolian also draws this distinction. The verb *med* in (19) is clearly used as an attitude verb. Second, although attitude verbs like *know* can embed definite descriptions under a concealed question reading (e.g., *Peter knows the time* means 'Peter knows what time it is'), such readings require the definite description to shift to an interrogative denotation. Thus, even if the *wh*-clause were analyzed as a definite description, it would still receive an interrogative interpretation. Finally, Mongolian lacks English-style free relatives (e.g., *John ate what Mary cooked*), meaning that a *wh*-constituent in Mongolian cannot function as a nominal argument of verbs like *eat*.

[5] Gong (2023) offers a syntactic analysis for the *gež*-type embedded questions, but leaves the other type open.

[6] In Mongolian, both *neg* and *negen* are used to form indefinites, but only *neg* is used for counting. Regarding *wh*-expressions, *negen* can follow most *wh*-expressions, whereas *neg* is typically used when the preceding *wh*-expression is *what*.

existence-asserting environments (Huang 1982; Cheng 1991; Li 1992; Lin 1998). However, Liu & Yang (2021), using naturally occurring data, argue that Mandarin *wh*-indefinites can also appear in various existence-asserting contexts. They propose analyzing these expressions as modal indefinites, following the approach of Alonso-Ovalle & Menéndez-Benito (2010), who suggest that all assertive sentences include a covert epistemic modal. As this debate does not impact the core argument of this paper, we will not explore it further here.

It is worth noting that, in addition to serving as a numeral, *neg(en)* can function as an indefinite determiner when combined with a common noun, as shown in (23). Unlike its use with *wh*-indefinites, *neg(en)* precedes common nouns rather than following them.[7]

(23) Bat delgüür-ees **neg nom** ab-san.
Bat.NOM store-ABL one book-ACC take-PST
'Bat bought a book at the store.'

The same pattern is consistently observed across conditionals, modals, and attitude verbs, all of which are established licensing environments of *wh*-indefinites in the literature. Concrete examples are listed below.

Conditionals. In Khorchin Mongolian, conditionals are marked by the morpheme *bol*, which typically form a constituent with antecedent clauses.[8] *Wh*-indefinites can occur within the antecedent clauses of conditionals.

(24) (Herbe) Bat **yuu neg** id-sen bol, inggež öls-h güi.
if Bat.NOM what one eat-PST COND such.a.way hungry not
'If Bat ate anything, he won't be hungry.'

(25) Bat önöödör **ali negen** tom od-iig har-san bol, lavtai hün
Bat.NOM today which one big star-ACC see-PST COND, definitely person
bolgon-d hel-ne.
every-DAT say-PRS
'If Bat saw any super star, then he would tell everyone about this.'

Modals. Both epistemic and deontic modals can license *wh*-indefinites.

(26) Bat **yuu neg** yum id-sen baiž magadgüi. Tegeed öls-h güi baina.
Bat.NOM what one thing eat-PST be may thus hunger-FUT not IS
'Bat may have eaten something. So, he won't be hungry.'

(27) Bat **yamar neg** yum id-h heregtei. Ügüi bol cusne čiher-ni
Bat.NOM what.kind one thing eat-INF must. not COND blood sugar-POSS

[7] This property parallels the numeral *yī* 'one' in Mandarin, which is also used to form ordinary indefinites. However, unlike Mongolian, *yī* is not required to co-occur with *wh*-indefinites in Mandarin.

[8] The conditional marker *bol* functions differently depending on its syntactic context. On one hand, it appears as an independent morpheme when it follows NPs, APs, or verbs with tense/aspect markers. On the other hand, it acts as a bound morpheme, written as *-bal* or *-bel*, complying with vowel harmony, when attached to a bare verb.

buur-na.
decrease-FUT
'Bat must eat something. Otherwise, he will get low blood sugar.'

Attitude verbs. *Wh*-indefinites can also be licensed in the complement clauses selected by attitude verbs. Note that the complementizer *gež* can also co-occur with *wh*-indefinites.

(28) Bat [biden-ig turšiltiin tasag-in **yuu neg**-ig ab-san gež]
Bat.NOM we-ACC experiment deparment-GEN what one-ACC take-PST COMP
sejigle-ž baina.
doubt-COV is
'Bat suspects that we have taken something from the laboratory.'

(29) Bat [biden-ig **ali negen** od-ig har-san gež] bod-san.
Bat.NOM we-ACC which one pop.star-ACC see-PST COMP think-PST
'Bat thought that we had seen a pop star.'

Polar questions. Our investigation shows that *wh*-indefinites also appear in polar questions, as illustrated in (30). In particular, the question particle *uu*, which marks polar questions, is incompatible with interrogative *wh*-expressions. Consequently, the presence of *uu* necessitates the use of *neg(en)*.[9],[10]

(30) Bat **yuu neg** id-sen uu?
Bat.NOM what one eat-PST PQ
'Did Bat eat something?'

Negation. It is important to note that in Mongolian *wh*-indefinites seem to be incompatible with negation, as shown in (31).

(31) *Bat **yuu neg** id-sen güi.
Bat.NOM what one eat-PST not
Intended 'Bat didn't eat anything.'

In a negative sentence, a non-interrogative *wh*-expression must precede the morpheme *č* 'too', rather than *neg(en)*, as demonstrated in (32). This pattern parallels the behavior of Mandarin universal *wh*-expressions, as shown in (33).

[9] One reviewer notes that their informants reject several examples like (24) and (30). As discussed earlier, dialectal variation is well attested. It is therefore possible that *wh*-indefinites cannot be licensed in polar questions in some Mongolian dialects, even though the construction is judged grammatical in Khorchin Mongolian—the dialect examined here. Alternatively, the acceptability of these examples may be due to contact with Mandarin, as our informants are bilingual in both Mandarin and Mongolian. These factors, while beyond the scope of the current study, are nevertheless of considerable interest and warrant further exploration.

[10] One reviewer considers the question particle *-uu* a clitic and proposes that it be attached to the sentence stem with a hyphen. Our view, however, is that the particle is better treated as an independent morpheme, comparable to sentence-final particles in Mandarin, which are not written in a clitic-like manner. Furthermore, previous studies on Mongolian questions, such as Onea & Guntsetseg (2011) and Gong (2023), also represent question particles as free morphemes.

(32) Bat **yuu č** id-sen <u>güi</u>.
Bat.NOM what too eat-PST not
'Bat didn't eat anything.'

(33) Libai **shenme ye** <u>mei</u> chi.
Libai what too not eat
'Libai didn't eat anything.'

Similar to the *wh*-expressions in (33), *wh*-expressions co-occurring with *č* are understood as universal free choice items. These expressions appear not only in negative environments, but also in other contexts that license free choice readings, such as modals (Wang 2025).[11]

One reviewer suggests that *neg(en)* 'one' may simply function as a specificity marker. If so, the infelicity of (31) would follow, since specific indefinites typically resist occurring in the scope of negation. While this line of reasoning is appealing, it fails to capture the broader distribution of Mongolian *wh*-indefinites. For instance, in modal and conditional environments, shown from (24) through (27), *wh*-indefinites clearly do not denote specific individuals in the speaker' s mind. Moreover, the cross-linguistic picture complicates the matter further: Mandarin *wh*-indefinites are not invariably non-specific under negation. In (34), for example, the most salient interpretation is an 'insignificant' reading, where the utterance conveys not that Libai ate nothing, but rather that the speaker knows exactly what he ate.

(34) Libai <u>mei</u> chi **shenme**. Jiu chi-le ge pingguo.
Libai not eat what just eat-PFV CLF apple
'Libai didn't eat much. He just had an apple.'

According to Huang (2013) and Chen (2021), the insignificant readings of *wh*-indefinites in Mandarin cannot be reduced to simple non-specificity. This suggests that the interaction between non-interrogative *wh*-expressions and negation cannot be fully explained in terms of specificity alone. Developing a comprehensive account of this interaction, however, lies beyond the scope of this paper and must be left for future research.

3.3. *Wh*-conditionals

We have established that *wh*-indefinites in Khorchin Mongolian must co-occur with the numeral item *neg(en)*, in contrast to interrogative *wh*-expressions, which require bare *wh*-forms. This distinction allows us to evaluate the classical variable-binding approach and the more recent question-based approach. Specifically, if the question-based approach is correct, the *wh*-expressions in *wh*-conditionals should align with those in *wh*-questions, meaning neither should

[11] In Mongolian, universal wh-expressions can also be formed by combining a *wh*-word with the particles *l* and *bol*, followed by a repetition of the wh-word, as illustrated below. Wang (2025) recently provided a semantic analysis of this construction.

(i) Bat hen-tei l bol hen-tei yari-dag.
Bat who-COM L COND who-COM talk-HAB
'Bat talks to anyone.' (Wang 2025: 1)

be compatible with *neg(en)*. This prediction is confirmed, as demonstrated in (35).

(35) {**Hen-ne** / ***Hen negen-ne**} surlag sain bol, {**hen** / ***hen negen**}
who-GEN who one-GEN learning good COND who.NOM who.NOM one
anggiin darag bol-no.
class monitor become-FUT
'For everyone x, if x has a good grade, x will be the class monitor.'

In Khorchin Mongolian, *wh*-conditionals are constructed similarly to their Mandarin counterparts: two *wh*-clauses are connected to form a conditional, marked by the conditional marker *bol*, with the *wh*-expressions co-referring. Unlike the *wh*-indefinites described in Section 3.2., these *wh*-expressions do not co-occur with *neg(en)*. Additional examples are provided below.

(36) Či **hežee** yab-bal, bi **hežee** ir-nee.
you.NOM when go-COND I.NOM when come-NPST
'For every time t if you leaves at t, I will come at t.'

(37) Bat **hen-ig** too-bal, Dorž **hen-ig** zal-naa.
Bat.NOM who-ACC like-COND Dorž.NOM who-ACC invite-NPST
'For everyone x, if Bat likes x, Dorž will invite x.'

(38) Ter **yaaž** Beejing-d yab-bal, bi **yaaž** Beejing-d yab-naa.
He.NOM how Beijing-DAT go-COND I.NOM how Beijing-DAT go-NPST
'No matter how you travel to Beijing, I will travel there the same way.'

(39) Či **yamar** šaltagaan-aar [detegber-d gar]-bal, bi bas **yamar** šaltagaan-aar
you what.kind reason-INS retire-COND I also what.kind reason-INS
[detegber-d gar]-na.
retire-NPST
'For everything x, if you retired because of x, I retired because of x.'

As in Mandarin, amount *wh*-expressions in Khorchin Mongolian are typically interrogative and do not have an indefinite use. This is evidenced by their incompatibility with the morpheme *neg(en)*, as illustrated in (40) and (41). However, amount *wh*-expressions can readily appear in *wh*-conditionals, as shown in (42).[12]

[12] The unacceptability of (41) may arise from a broader incompatibility between *neg(en)* and numerals. As discussed in Section 3.2., *neg(en)* literally means 'one', which naturally conflicts with numerals greater than one. This incompatibility is illustrated in the following example, where the *wh*-indefinite fails to co-occur with *gurban* 'three' :

(i) *Baatar **ali** **negen** gurban nom-ig uns-san bololtoi.
Baatar.NOM which one three book-ACC read-PST probably
Intended 'Baatar may have read three books.'

In Khorchin Mongolian, plural *wh*-indefinites are typically formed with the *wh*-item *hedun*, which can be treated as the counterpart of Mandarin *ji* 'several/how many', as shown in (ii). Like *ji*, *hedun* can also be interpreted interrogatively.

(ii) Baatar **hedun** **nom** unš-san bololtoi.
Baatar.NOM how.many book read-PST probably
'Baatar may read several books.'

(40) Baatar **hedii** **nom** ab-san be?
Baatar.NOM how.many book buy-PST Q
'How many books did Baatar buy?'

(41) *Baatar **hedii** **negen** nom ab-san.
Baatar.NOM how.many one book buy-PST
Intended 'Baatar bought some book.'

(42) Baatar **hedii** **nom** hudald-bal, bi bas **hedii** **nom** ab-na.
Baatar.NOM how.many book sell-COND I also how.much book buy-NPST
'For every amount n, if Baatar sells n books, I would buy n books.'

Moreover, in Khorchin Mongolian, while negation is incompatible with *wh*-indefinites (see Section 3.2.), it occurs naturally in *wh*-questions, as shown in (43). If the *wh*-clauses in a *wh*-conditional pattern more closely with interrogative clauses, the presence of negation in *wh*-conditionals is expected. This prediction is borne out by (44).

(43) Baatar **yuu** id-h güi be?
Baatar what eat-INF not Q
'What does Baatar not eat?'

(44) Baatar **yuu** id-h güi bol, bi bas **yuu** ide-h güi.
Baatar what eat-INF not COND I also what eat-INF not
'For every food x, if Baatar does not eat x, I will not eat x.'

These data suggest that *wh*-expressions in Mongolian *wh*-conditionals align more closely with those in *wh*-questions than with non-interrogative *wh*-indefinites. This pattern is naturally accounted for under the question-based approach, which treats *wh*-expressions in *wh*-conditionals as inherently interrogative. Hence, it is expected that they share the same properties with those in *wh*-questions.

However, this argument is not decisive. The variable-binding approach can respond by positing that *neg(en)* functions as a local binder for the variable introduced by a *wh*-expression, while the question operator and the necessity modal in conditionals serve as global binders. More specifically, *neg(en)* may be analyzed as an overt DP-level existential operator that converts a *wh*-expression into an existential quantifier. This analysis also aligns with the exhaustification approach to *wh*-indefinites, which accounts for their licensing conditions in terms of the existential force contributed by the *wh*-expressions themselves (Liao 2011; Chierchia & Liao 2015; Liu & Yang 2021, among others). Once a *wh*-word combines with *neg(en)*, it no longer introduces a variable, which explains why *neg(en)* is incompatible with both *wh*-questions and *wh*-conditionals. In this respect, *wh*-indefinites differ fundamentally from the other two *wh*-constructions, as their interpretation does not involve long-distance binding. While the incompatibility between *neg(en)* and negation- or amount-based *wh*-expressions remains unresolved, it may reflect certain as-yet-unknown constraints on the existential quantifiers derived from *wh*-

However, it is worth noting that *hedun*, unlike other *wh*-words, is incompatible with *neg(en)*. Accounting for this difference lies beyond the scope of the present paper and is left for future research.

expressions. Under this view, *wh*-expressions in *wh*-conditionals can still be treated as variables that are never interpreted interrogatively at any point in the derivation.

To strengthen the argument, the reminder of this section demonstrates that *wh*-conditionals exhibit properties that are exclusively associated with interrogative clauses.

Focus-related properties: Intervention and association. In languages that allow both interrogative and existential interpretations of *wh*-expressions, only the interrogative use interacts with focus, whereas the existential use does not. It is widely recognized that *wh*-expressions typically serve as the primary focus in interrogative sentences, while *wh*-indefinites are generally not associated with focus (Dong 2009, 2018; Haida 2007; Trukenbrodt 2013; Yang 2018; Hengeveld et al. 2023, among others). Although Mongolian lacks a dedicated morpheme for marking focus, focusability can be diagnosed through focus association. In particular, like many other languages, Mongolian has a set of exclusive particles that are sensitive to focus. As demonstrated in (45-a) and (45-b), the particles *l* and *gagčhan* in Khorchin Mongolian function similarly to the English *only*, highlighting that the focused constituent associating with them uniquely satisfies the relevant predicate.[13] Throughout this paper, focus in example sentences is indicated using italics.

(45) a. Baatar *ene nom*-ig l unšž öngör-žee.
Baatar.NOM this book-ACC only read.COV pass-PST
b. Baatar gagčhan *ene nom*-ig unšž öngör-sen.
Baatar.NOM only this book-ACC read.COV pass-PST
'Baatar read only this book.'

In *wh*-questions, these focus-sensitive particles can take *wh*-expressions as their focus associates, as illustrated in (46) and (47). In contrast, *wh*-indefinites cannot function as focus associates, as shown in (48) and (49).

(46) a. Baatar **hen-ig** l uri-h be?
Baatar.NOM who-ACC only invite-PST Q
b. Baatar gagčhan **hen-ig** uri-h be?
Baatar only who-ACC invite-PST Q
'Who was the person x such that Baatar invited only x?'

(47) a. Baatar **yamar nom-ig** l unšž öngör-sen be?
Baatar.NOM what.kind book-ACC only read pass-PST Q
b. Baatar gagčhan **yamar nom-ig** unšž öngör-sen be?
Baatar.NOM only what.kind book-ACC read pass-PST Q
'What was the kind of book x such that Baatar read only x?'

[13] Literally speaking, *gagčhan* is closer in meaning to *sole* or *alone*, emphasizing singularity and typically appearing in more formal registers. In contrast, *l* is multifunctional: in addition to marking exclusivity, it can also convey meanings akin to *still* or *right after*. It is more frequently used in colloquial Mongolian. As the fine-grained semantic differences between these exclusive particles are not central to the present analysis, we set them aside in what follows.

(48) a. *Baatar **hen negen** l uri-san.
Baatar.NOM who one only invite-PST

b. *Baatar gagčhan **hen negen** uri-san.
Baatar only who one invite-PST
Intended 'Baatar invited only someone.'

(49) a. *Baatar **yamar negen nom-ig** l unšž öngör-sen.
Baatar.NOM what.kind one book-ACC only read.COV pass-PST

b. *Baatar gagčhan **yamar negen nom-ig** unšž öngör-sen.
Baatar.NOM only what.kind one book-ACC read.COV pass-PST
Intended 'Baatar read only some kind of book.'

The infelicity of (48) and (49) may be due to their non-informativeness. Exclusive particles typically introduce an existential presupposition while asserting exhaustivity (e.g., Rooth 1992; Tsai 2004; Coppock & Beaver 2014). For instance, in (45-a) and (45-b), it is presupposed that there was a book Baatar read, and the sentences assert that he read only that book. In the case of (48) and (49), the use of *wh*-indefinites yields existential statements, thereby asserting what is already presupposed and offering no new information. In contrast, the *wh*-questions in (46) and (47) seek to identify the unique individuals, in addition to presupposing their existence.

If *wh*-expressions in a *wh*-conditional are interrogative, they are expected to be focused as well. This prediction is confirmed by the following examples, where the exclusive particles are associated with the *wh*-expressions.

(50) a. Baatar **hen-ig** l uri-bal, bi bas **hen-ig** l uri-na.
Baatar.NOM who-ACC only invite-COND I.NOM also who-ACC only invite-NPST

b. Baatar gagčhan **hen-ig** uri-bal, bi bas gagč **hen-ig**
Baatar.NOM only who-ACC invite-COND I.NOM also only who-ACC
uri-na.
invite-NPST
'For everyone x, if Baatar invites only x, then I also invites only x.'

(51) a. Baatar **yamar nom-ig** l unš-bal, bi tuun-d **yamar nom-ig**
Baatar.NOM what book-ACC only read-COND I.NOM he-DAT what.kind book
l ög-nö.
only give-NPST

b. Baatar gagčhan **yamar nom-ig** unš-bal, bi tuun-d gagčhan
Baatar.NOM only what book-ACC read-COND I.NOM he-DAT only
yamar nom ög-nö.
what.kind book give-NPST
'For every kind of book x, if Baatar reads only x, I will give only x to him.'

In addition to focus association, another interaction of interrogative *wh*-expressions and focus is manifested through a phenomenon known as 'focus intervention effects'. In particular, focused elements associated with focus-sensitive particles are prohibited from preceding interrogative *wh*-expressions, as illustrated in the following examples. The unacceptable sentences in (52-a) and (53-a) become acceptable when the *wh*-expression is scrambled to precede the focus element, as shown in (52-b) and (53-b).[14]

(52) a. **Baatar* l **yuu-ig** ab-san be?
Baatar.NOM only what-ACC buy-PST Q

b. **Yuu-ig** *Baatar* l ab-san be?
what-ACC Baatar.NOM only buy-PST Q
'Who was the person x such that only Baatar met x?'

(53) a. *Gagčhan *Baatar* **yuu-ig** ab-san be?
only Baatar.NOM what-ACC buy-PST Q

b. ?**Yuu-ig** gagčhan *Baatar* ab-san be?
what-ACC only Baatar.NOM buy-PST Q
'Who was the person x such that only Baatar met x?'

By contrast, focus intervention effects typically do not arise in declarative sentences.[15] As a result, exclusive particles and their focus associates can precede *wh*-indefinites without causing unacceptability, as illustrated in (54).

(54) a. *Baatar* l **yuu neg** ab-san (bololtoi).
Baatar.NOM only what one buy-PST probably

b. Gagčhan *Baatar* **yuu neg** ab-san (bololtoi).
only Baatar.NOM what one buy-PST probably
'Only Baatar (may have) bought something.'

However, focus intervention effects do arise in *wh*-conditionals, as illustrated in (55) and (56). Naturally, these effects disappear when the *wh*-expression precedes the focus elements, as shown in (57).

(55) a. *Gagčhan *Baatar* **hen-ig** uri-bal, bi **hen-ig** uri-na.
only Baatar.NOM who-ACC invite-COND, I who-ACC invite-NPST

[14] Branan (2018) observes that not all focus-sensitive particles and their associates intervene in Mongolian. For example, the exclusive particle *zövxöng* appears not to cause such effects, according to Branan, as illustrated below.

(i) Zövxöng *uran* *noyon* **yamar** üneer-g zursan be?
only artistic knight what cow-ACC paint Q
'What cow did only the artistic knight draw?' (Branan 2018: 235)

However, this sentence was judged unacceptable by our consultants, and the use of *z” ovx” ong* is rare in spoken Mongolian. The source of this discrepancy in acceptability is unclear. While dialectal variation may be a contributing factor, this issue warrants further investigation.

[15] Li & Law (2016) argues that focus intervention effects may appear to interfere with the interpretations of *wh*-indefinites and disjuctive expressions in Mandarin and English. However, their analysis is framed within alternative semantics, where *wh*-indefinites and disjunctive expressions share the same semantic type as interrogative *wh*-expressions—namely, sets of alternatives—until they met appropriate closure operators. Further discussion of this analysis can be found in Section 4.. Crucially, focus intervention effects arise only within the scope of such closure operators. In this sense, the effects remain tied to interrogative-style denotations, that is, alternative sets.

b. **Baatar* l̲ **hen-ig** uri-bal, bi **hen-ig** uri-na.
Baatar.NOM only who-ACC invite-COND, I who-ACC invite-NPST
Intended 'For everyone x, if only Baatar invites x, I will invite x.'

(56) a. *Baatar **yamar** **nom** bič-bel, gagčhan *tednüs* **yamar** **nom** hebel-ne.
Baatar.NOM what.kind book write-COND, only they what.kind book publish-NPST

b. *Baatar **yamar** **nom** bič-bel, *tednüs* l̲ **yamar** **nom** hebel-ne.
Baatar.NOM what.kind book write-COND, they only what.kind book publish-NPST
Intended 'For every kind of book x, if Baatar writes x, only they will publish x.'

(57) a. **Ali** **kompani** gagčhan *šinehen tögsögči-d-ig* else-bel, **ali** **kompani-d** tobč namtar-aa ög.
which company.NOM only fresh graduate-PL-ACC recruit-COND, which company-DAT resume-RX give

b. **Ali** **kompani** *šinehen tögsögči-d-ig* l̲ else-bel, **ali** **kompani-d** tobč namtar-aa ög.
which company.NOM fresh graduate-PL-ACC only recruit-COND, which company-DAT resume-RX give
'For every company x, if x only recruits recent graduates, you should apply to x.'

Focus intervention effects are widely attested in *wh*-questions. While various accounts have been proposed to explain this phenomenon, they consistently attribute it to the syntactic (e.g., Beck 1996; Kim 2006; Yang 2012; Li & Cheung 2015), semantic (e.g., Beck 2006; Mayr 2014; Kotek 2014; Li & Law 2016; Erlewine 2025), or pragmatic (e.g., Tomioka 2007) properties of interrogative *wh*-expressions. Therefore, if focus intervention effects are observed in the *wh*-clauses of *wh*-conditionals, the *wh*-expressions involved must likewise possess interrogative features.

In short, only interrogative-denoting elements exhibit both focus association and focus intervention effects. The striking parallels between *wh*-conditionals and *wh*-questions are thus expected if the former are derived from the latter.

Mention-some interpretations. Under the variable-binding approach, *wh*-conditionals yield universal quantification by default, as the variables introduced by *wh*-expressions are bound by the necessity operator in conditionals. For instance, (58) is true if and only if Dorž will invite **all** the people that Bat likes.

(58) Bat **hen-ig** too-bal, Dorž **hen-ig** zal-naa.
Bat.NOM who-ACC like-COND Dorž.NOM who-ACC invite-NPST
'For everyone x, if Bat likes x, Dorž will invite x.'

While this universal quantification-based truth condition captures most cases, a *wh*-conditional can receive an existential interpretation when its antecedent clause is construed as a mention-some *wh*-question (Xiang 2016, 2020; Liu 2016, 2017). This observation is also confirmed in Mongolian. Consider (59), which infers that the speaker will go to one (but not necessarily all) of the places which sold milk tea.

(59) Haa süü-tei-čee hundaldaž bai-bal, bi haana oči-no.
where milk-tea sell-COV be-COND I.NOM go-NPST
'I will go to a place which sold milk tea.'

The contrast between (58) and (59) corresponds to a well-known distinction between 'mention-all' and 'mention-some' interpretations of *wh*-questions. When used as a question, the antecedent *wh*-clause in (58) calls for a mention-all answer—that is, an appropriate response must enumerate all the individuals Bat likes. In the following conversation, for example, the answer gives rise to the inference that Bat likes only the individuals who are explicitly mentioned.

(60) A: Bat **hen-ig** too-h be?
Bat.NOM who-ACC like-INF Q
'Who does Bat like?'
B: Nabči, Ceceg, Naran-ig too-no.
Nabči Ceceg Naran-ACC like-NPST
'He likes Nabči, Ceceg, and Naran.'

However, the antecedent *wh*-clause in (59) receives a mention-some interpretation when used as a question. For instance, in (61), Speaker A asks the question to find out where milk tea can be purchased. Speaker B provides one possible location, but this response does not imply that it is the only place where milk tea is available.[16]

(61) A: Haa süütei-čee hundaldaž bai-h be?
where milk-tea sell-COV be-INF Q
'Where is milk tea being sold?'

[16] The mention-some interpretation is often associated with the presence of modals. A well-known example illustrating this reading is the question *Where can I buy coffee?* In Mongolian, modal questions similarly yield a mention-some interpretation, as demonstrated in (i). However, based on our investigation, such questions are not frequently used in everyday conversation.

(i) **Haana-aas** süütei-čee hudaldan-ab-z <u>diileh</u> be?
where-ABL milk-tea buy-COV can Q
'Where can I buy milk tea?'

In Mandarin as well, a question with the same form as (61) leads to a mention-some interpretation, as shown in (ii).

(ii) **Nǎlǐ** yǒu jiàngyóu mài?
where have soy.sauce sell
'Where is soy sauce being sold?'

While we acknowledge that modals play a central role in existing semantic accounts of mention-some interpretations (George 2011; Fox 2012; Xiang 2022), Van Rooy (2004) proposes a pragmatic approach that derives such interpretations from the contextual utility of questions. Given that the examples in (61) and (ii) do not involve modals, the pragmatic account may offer a better explanation in these cases.

B: Asar doorah ter delgüür-t bai-na.
downstairs that store-DAT be-NPST
'At the store downstairs.'

The interpretative parallel between *wh*-conditionals and *wh*-questions is explicable under the assumption that the former are semantically derived from the latter.

The question particle. Before concluding this section, we highlight a key difference between *wh*-conditionals and *wh*-questions. As shown in Section 3.1., *wh*-questions in Mongolian consistently require the question particle *be*, which also appears in embedded *wh*-questions with the complementizer *gež*. The relevant data are repeated below for ease of reference.

(62) Bat **yuu** id-sen be?
Bat.NOM what eat-PST Q
'What did Bat eat?'

(63) Bat Dorž-oos [**yuu** id-sen be gež] asuu-san.
Bat.NOM Dorž-ABL what eat-PST Q COMP ask-PST
'Bat asked Dorž what he ate.'

However, *be* is not permitted in *wh*-conditionals. This raises a puzzle: if *wh*-conditionals are composed of two interrogative *wh*-clauses, why is the question particle absent? While we do not resolve this issue in the present paper, we maintain that the absence of *be* does not entail that the *wh*-clause is non-interrogative. First, not all embedded *wh*-questions can take the question particle *be*. As shown in (64), embedded questions that lack the complementizer *gež* also disallow the presence of *be*. This suggests that the occurrence of *be* is conditioned by the presence of *gež* and is not a necessary component of embedded *wh*-questions. In Section 4., we offer a semantic definition of the question particle within the framework of alternative semantics, where interrogative meaning originates from *wh*-expressions rather than from a dedicated question operator.

(64) Bat [Dorž-in **yuu** id-sen-ig (*be)] asuu-san.
Bat.NOM Dorž-GEN what eat-PST-ACC Q ask-PST
'Bat asked what Dorž ate.'

Additionally, in a conditional construction, both clauses are subordinate but not selected by attitude verbs. This structural configuration may prevent the use of the complementizer *gež*, and consequently, the question particle *be* is also disallowed. In other words, the incompatibility of *be* in *wh*-conditionals may stem from the structural exclusion of *gež* in conditionals, rather than from the *wh*-clauses themselves not being interrogative.

Thus, the distribution of the question particle may largely be determined by structural factors. Under the cartographic approach (Rizzi 1997), the question particle may serve as a marker of the Force head in the left periphery (see also Gong 2023), which appears only in matrix clauses or in fully projected embedded clauses selected by the complementizer. This hypothesis, of course, calls for further investigation from a syntactic perspective.

3.4. Interim summary

Based on the comparison among *wh*-indefinites, *wh*-conditionals, and *wh*-questions in Khorchin Mongolian, we can conclude that *wh*-expressions in a *wh*-conditional exhibit many properties only available in *wh*-questions. In this light, an analysis that accounts for the interrogative nature of *wh*-conditionals is more appropriate. In the following section, we offer such an analysis.

4. A question-based analysis of *wh*-conditionals

The generalization reached in Section 3.3. supports a heterogeneous view of non-interrogative *wh*-constructions: although they do not ultimately serve an interrogative function, their underlying sources may differ. According to the question-based approach, *wh*-conditionals originate from interrogative *wh*-clauses. Specifically, a *wh*-conditional is formed by linking two clauses that each denote an interrogative meaning. As noted in the introduction, *wh*-conditionals can be interpreted as encoding dependencies between question resolutions, a relationship captured by the interpretive schema in (65).

(65) $[\![$COND WH-clause$_1$, WH-clause$_2]\!]$ = for every world w and object x, if x resolves the issue raised by WH-clause$_1$ in w, then x resolves the issue raised by WH-clause$_2$ in w.

In (65), both WH-clause$_1$ and WH-clause$_2$ are interpreted as interrogative clauses that introduce issues. Together, they form a conditional statement by expressing that the resolution of the issue raised by WH-clause$_2$ depends on the resolution of the issue raised by WH-clause$_1$. This dependency is readily observable in a common use of *wh*-conditionals—namely, as a strategy for addressing questions under discussion, as illustrated in the conversation in (66).

(66) A: Či **hežee** ir-sen be?
you when come-PST Q
'When did you come?'

B: Či **hežee** yab-bal, bi **hežee** ir-nee.
you when go-COND I when come-NPST
'For every time t if you leaves at t, I will come at t.'

Given that both clauses in a *wh*-conditional are interrogative in nature, it is unsurprising that *wh*-conditionals exhibit properties typically associated with interrogative clauses.

In contrast, *wh*-indefinites denote non-interrogative existential quantifiers. This is consistent with the exhaustification approach, in which *wh*-indefinites are typically analyzed as existential in nature. As suggested in Section 3.3., the morpheme *neg(en)* in Khorchin Mongolian functions as an overt DP-level marker signaling this non-interrogative use. Hence, sentences with *wh*-indefinites are inherently declarative, rather than derived from interrogative clauses. This distinction helps explain their sharp contrast with *wh*-questions and *wh*-conditionals.

While a comprehensive account of the relationships among these *wh*-constructions lies beyond the scope of this paper, sketching a potential analytical direction may offer a useful foundation for future research. In the remainder of this section, we present a semi-formal discussion grounded in the alternative semantics approach to questions. We do not advocate for this particular framework here; rather, we adopt it as a concrete tool to facilitate exposition. The core ideas developed below can be adapted to other theoretical frameworks at the reader's discretion.

4.1. A semi-formal analysis based on alternative semantics

We adopt the general framework of interrogative meaning proposed by Hamblin (1973), under which a question denotes a set of propositions. Various compositional approaches have been developed to derive this semantic representation, and a critical overview of these approaches can be found in Dong (2009), Lin (2014), or Li & Shi (2022). In this paper, we situate our discussion within the alternative semantics approach. A primary motivation for this choice is that many current analyses of focus-related properties of interrogative *wh*-expressions—such as focus intervention and focus association (see Section 3.3.)—are formulated within this framework (e.g., Beck 2006; Eckardt 2007; Kotek 2014; Li & Law 2016; Erlewine 2025). Since these phenomena play a crucial role in our argument, adopting alternative semantics thus allows them to be accounted for in a unified and theoretically consistent manner.[17]

In the alternative semantics approach, *wh*-expressions are assumed to denote a set of alternatives, as in (67), which combines with other lexical items in a point-wise manner, generating a set of propositions at the root of a *wh*-clause, as shown in (69). In particular, the derivation is facilitated by two polymorphic functions—η and $\circledast$—in the sense of Charlow (2017), defined as in (68) (a and b stands for variables of types). The function η trivially shifts a semantic value to a singleton set containing this value, while the function $\circledast$ embodies a pointwise compositional rule. It takes two sets, F and X, combines one member from F with another one from X, and exhausts all combinations.[18]

(67) $[\![\text{yuu}_u]\!]^g = \{x \mid x \in \mathsf{thing}_{w_0}\}$ Type: $\mathsf{e} \to \mathsf{t}$

(68) Combinatorial rules in a point-wise manner

a. $\eta(x) = \{x\}$ Type: $a \to (a \to \mathsf{t})$

b. $F \circledast X = \{f(x) \mid f \in F, x \in X\}$ Type: $((a \to b) \to \mathsf{t}) \to (a \to \mathsf{t}) \to (b \to \mathsf{t})$

(69) $$\begin{aligned}[\![\text{Bat yuu id-sen}]\!]^{w_0} &= (\eta[\![\text{Libai}]\!]) \circledast ([\![\text{yuu}]\!]^{w_0} \circledast (\eta[\![\text{id-sen}]\!])) \\ &= (\eta[\![\text{Bat}]\!]) \circledast \{\lambda y \lambda w.\mathsf{ate}_w(x)(y) \mid x \in [\![\text{yuu}]\!]^{w_0}\} \\ &= \{\lambda w.\mathsf{ate}_w(\mathsf{b}, x) \mid x \in \mathsf{thing}_{w_0}\}\end{aligned}$$

In this approach, the default semantic contribution of a *wh*-constituent is interrogative. Hence,

[17] That said, alternative-based analyses can be broadly divided into two camps. For instance, Beck (2006) proposes a unified framework that treats focus and *wh*-alternatives within a single system, a view further developed by Kotek (2014) and Erlewine (2025). In contrast, Eckardt (2007) and Li & Law (2016) adhere to a more traditional interpretation of alternative semantics and advocate for a non-unified approach. Nonetheless, this theoretical distinction is not central to the issues addressed in this paper.

[18] The semantic types used in this paper include: e (individuals), t (truth values), s (possible worlds), and g (assignments). Functional types are represented using the schema $a \to b$ for any type a and b. For simplicity, intensional types in the form $\mathsf{s} \to a$ are abbreviated as $\mathsf{s}a$.

the Question operator is trivial in the at-issue component, but it may impose a definedness condition—specifically, that the set of propositions it operates on be non-singleton, as formalized in (70). In Mongolian, the particle *be* may function as an overt realization of this operator.

(70) $[\![\mathbb{Q}/\text{be}]\!] = \lambda \mathcal{Q}.\mathcal{Q}$ defined only if $|\mathcal{Q}| > 1$ Type: $(\mathsf{st} \to \mathsf{t}) \to (\mathsf{st} \to \mathsf{t})$

Given the default interrogative interpretation of *wh*-constituents, the presence of a question particle is not required to derive interrogative meaning. This is in line with the observation that such particles do not consistently appear in embedded environments (see Section 3.1.).

For the same reason, *wh*-indefinites, rather than interrogative *wh*-expressions, are expected to bear distinct morphological marking. In Mongolian, the morpheme *neg(en)* 'one' can be analyzed as an overt DP-level Existential Closure, defined in (71). Formally, it is a function that shifts a *wh*-expression into an existential quantifier. As a result, the *wh*-expression contributes no alternatives after combining with the Existential Closure, and the sentence containing a *wh*-indefinite yields a proposition as its meaning.

(71) $[\![\text{neg(en)}]\!] = \lambda \mathcal{S} \lambda P \lambda w \exists x.x \in \mathcal{S} : P(x)(w)$ Type: $(\mathsf{e} \to \mathsf{t}) \to (\mathsf{e} \to \mathsf{st}) \to \mathsf{st}$

Consequently, when combined with *neg(en)*, a *wh*-expression is shifted to an existential quantifier, which underlies the non-interrogative interpretation in many analyses of *wh*-indefinites (Liao 2011; Law 2018; Liu & Yang 2021; Hengeveld et al. 2023). The definition can be justified independently by the fact that *neg(en)* serves as an indefinite determiner when combining with a common noun (see Section 3.2.).[19] This paper does not offer a thorough discussion of how an existential quantifier denotation interacts with modals or conditionals to produce the various interpretive properties of *wh*-indefinites. Readers interested in these interactions are encouraged to consult the referenced studies (Liao 2011; Chierchia & Liao 2015; Law 2018; Liu & Yang 2021; among others).

In the case of *wh*-conditionals, the *wh*-expressions appear in their bare form, identical to interrogative *wh*-forms. As a result, the *wh*-clauses naturally receive interrogative interpretations through the compositional process. According to the question-based approach, the interrogative interpretations can be transformed through a non-clausal answerhood condition. In particular, the *wh*-clauses involved are interpreted interrogatively and raise issues. Importantly, these issues can be resolved by non-clausal objects, as evidenced by the fact that answers to *wh*-questions may be non-clausal, shown in the following examples.

[19] Recall (23). When combining with a common noun, *neg(en)* precedes it, rather than following it, contrasting with its distribution in forming *wh*-indefinites. This difference suggests that the derivation of a *wh*-indefinite is not identical to that of an ordinary indefinite. While the present definition of *neg(en)* cannot fully account for this distinction, it does not predict that *neg(en)* should combine with a common noun in the same manner as with a *wh*-expression. In brief, a common noun is conventionally modeled as an intensional individual predicate of type $\mathsf{e} \to \mathsf{st}$, which cannot serve as input to the function defined in (71). Providing a comprehensive account for this distributional difference falls beyond the scope of this paper and is left for future research.

(72) A: Who did Bat invite?
B: Dorž.

(73) A: When did Bat arrive?
B: 6 pm.

While there is ongoing debate over whether such non-clausal responses (alse known as 'fragments') are derived syntactically or semantically, Chierchia (2013), Xiang (2020), and Cremers (2018) have independently argued that non-clausal answerhood plays a crucial role in the semantics of free relatives and embedded *wh*-questions. Without delving into formal details, we adopt the following semi-formal answerhood condition, which relies on Li's (2021) specific implementation, to guide the discussion.[20] We also include several essential formal definitions in Appendix A to help interested readers understand how interrogative meaning can be transformed into a predicate meaning.

(74) $\textbf{SA}[\![\text{who did Bat invite}]\!] = \lambda x \lambda w.$ x is in the domain of *who* and x is an individual covering all people invited by Bat in w.

Basically, **SA** is a operator mapping an interrogative meaning to a predicative meaning. The latter can serve as the argument of the quantificational operator introduced in a conditional. Following the Lewis–Kratzer–Heim approach to conditionals (Lewis 1975; Kratzer 1981, 1986; Heim 1982), the meaning of a conditional is not the material implication in first order logic, but a quantification over possible worlds (or situations) triggered by a modal (e.g., *must*, *may*, *can*, etc.) or a quanticational adverb (e.g. *always*, *usually*, *sometimes*, etc.) in the clause B. When no such lexical items appear in a conditional, a covert necessity modal is posited to produce the default necessity statement. These quantifiers quantify over possible worlds as well as other variables when they are available, i.e., they unselectively bind variables (see also Chierchia 1992, 2000). In (75), $\mathfrak{M}$ is a quantificational operator, and $[\![A]\!]$ and $[\![B]\!]$ denote functions taking a world argument as well as an argument sequence $\vec{x}$ of any length including 0 to a proposition.

(75) $[\![\textsc{cond}\ A, B]\!]_d = \mathfrak{M}_{\vec{x},w}[[\![A]\!]_d(\vec{x})(w)] : [\![B]\!](\vec{x})(w)$

As an illustration, consider (77). Since there is no overt modal included, the meaning of the *wh*-conditional is a quantification evoked by a covert necessity modal $\mathbb{NEC}$. The restrictor and scope of $\mathbb{NEC}$ are saturated by functions yielded by applying **SA** to *wh*-clauses. Thus, (77) can be interpreted as (77).

(76) Bat yuu id-sen bol, Dorž yuu id-sen.
Bat.NOM what eat-PST COND Dorž.NOM what eat-PST
'For everything x, if Bat ate x, Dorž ate x.'

(77) $[\![\mathbb{NEC}]\!]_{x,w'}[\textbf{SA}[\![\text{Bat yuu ate}]\!](x)(w')] : \textbf{SA}[\![\text{Dorž yuu ate}]\!](x)(w')$

$= \lambda w \forall x, w'. w' \in \text{ACC}(w) \wedge [\![\text{Bat yuu ate}]\!](x)(w') : [\![\text{Dorž yuu ate}]\!](x)(w')$

In prose: $\lambda w.$ for all x and w' that is accessible to w, if x is an individual covering all

[20] It is worth noting that Xiang (2016, 2020) and Liu (2016, 2017) offer different implementations of the question-based approach. The distinctions between these proposals are not directly relevant to the present discussion, but interested readers may consult Li (2021) for a comparative overview.

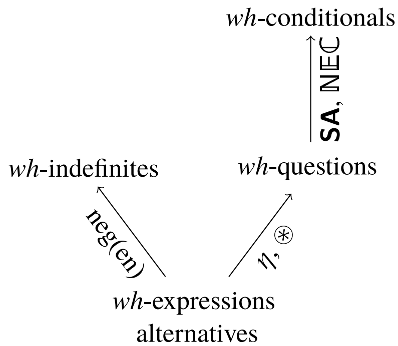


Figure 2: Question-based approach (final)

things that Bat ate, then x is also an individual covering all things that Dorž ate.

The resulting interpretation provides a concrete instance of the general interpretive schema in (65). By means of the non-clausal answerhood operator, the two interrogative *wh*-clauses are linked to yield a non-interrogative conditional meaning.

Thus, the present analysis provides an explicit framework for deriving *wh*-questions, *wh*-indefinites, and *wh*-conditionals from alternatives. Within this framework, wh-conditionals are based on wh-questions and are independent of *wh*-indefinites. Therefore, Figure 1b can be refined and detailed as shown in Figure 2.

4.2. Question-based approach vs. Variable-binding approach

While the question-based approach fundamentally differs from the variable-binding approach, this does not preclude it from incorporating some of the latter's advantages. In the variable-binding approach, *wh*-expressions uniformly introduce free variables, whose values remain unspecified until bound by an appropriate operator. By contrast, the question-based approach assigns fixed semantic values to *wh*-expressions, eliminating the need for variables. Nonetheless, binding remains relevant within the question-based framework. As defined in (75), conditionals involve an operator that unselectively binds variables introduced in the predicates derived from the interrogative meanings of *wh*-clauses. This mechanism of unselective binding is, in fact, central to the variable-binding approach. In this respect, the question-based approach can be seen as introducing a semantic shifter that effectively 'creates' variables for the conditional operator to bind. The relation between these two approaches can be depicted in Figure 3. Consequently, it not only accounts for the interrogative properties observed in *wh*-conditionals but also retains a key component of the variable-binding approach.[21]

Building on the connection between the variable-binding and question-based approaches,

[21] A similar connection is found in Chierchia's (2000) analysis of *wh*-conditionals. In his approach, *wh*-expressions uniformly denote non-interrogative existential quantifiers. However, within the framework of dynamic semantics, existential quantifiers can be converted into variables (see Appendix A), thereby reintroducing the core idea of unselective binding.

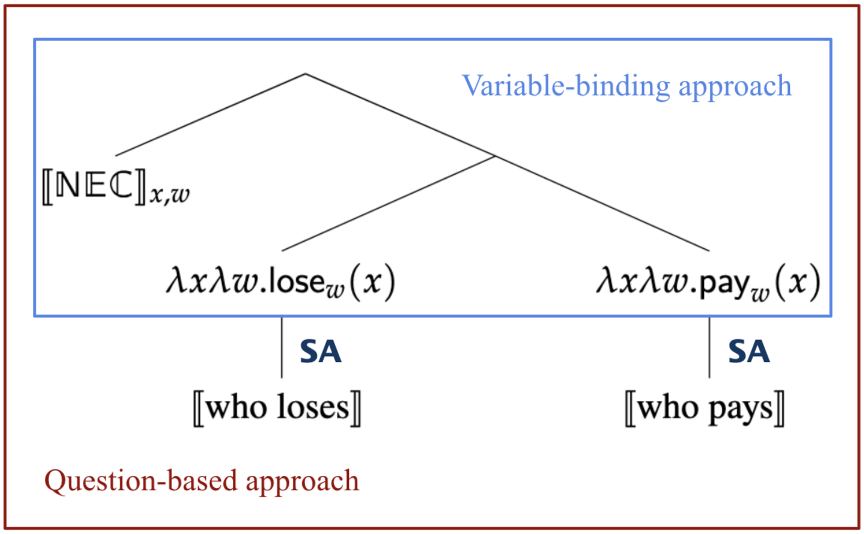


Figure 3: Question-based approach vs. Variable-binding approach

the latter can also account for the morphological identity constraint on *wh*-expressions in *wh*-conditionals, as noted by Cheng & Huang (1996). In the Mandarin example in (78), although both *na-ge nianqingren* and *na-ge qingnian* mean 'which youth', they are not interchangeable in the consequent clause. Intuitively, if the second *wh*-expression is *na-ge qingnian*, it cannot refer back to the same individual introduced by the first expression. This morphological matching requirement is also observed in Mongolian.

(78) Na-ge nianqingren shu-le, na-ge {nianqingren / *qingnian} jiu qingke.
which-CL youth lose-ASP which-CL youth youth then pay
'For every youth x, if x is the one losing the bet, x is the one paying.'

In the variable-binding approach, this constraint is attributed to two variables being directly and simultaneously bound by the same operator. Under the question-based approach, since interrogative *wh*-expressions also introduce variables, the same explanation applies naturally to account for this constraint.

Moreover, the connection of these two approaches allows us to respond to Tsai's (2023) critique of the question-based approach. Tsai observes that, in Mandarin, *wh*-conditionals are correlated with another construction known as 'reflexive doubling', as illustrated in (79). In this construction, two clauses—each containing a co-indexed reflexive—are conjoined to express a conditional meaning.[22]

[22] This construction cannot simply be analyzed to contain two clauses with empty subjects. (79) becomes unacceptable once reflexives are dropped, as shown in (i). It indicates the necessity of co-indexed reflexives in the formation of this construction.

(i) *Fàn-le cuò, jiù chùlǐ.
make-PFV mistake just handle

(79) **Ziji** fan-le cuo, jiu **ziji** chuli.
self make-PFV mistake just self handle
'For everyone x, if x himself makes a mistake, x must handle it by himself.'

Clauses containing reflexives cannot be analyzed as interrogative, but reflexives are typically treated as variables. As such, the variable-binding approach can straightforwardly extend its binding mechanism from *wh*-conditionals to reflexive doubling constructions. In contrast, the question-based approach may not naturally account for such cases.

A similar concern arises with the following kind of construction noted in Wen (2006) and Pan & Kuang (2023). In this example, the two clauses contain indefinites that appear to be co-referential, and their juxtaposition yields a conditional interpretation. The absence of *wh*-expressions precludes a question-based analysis.

(80) **Yi-ge ren** zuo shi, **yi-ge ren** dang.
one-CL person do thing one-CL person be.responsible
'For everyone x, if x has done something wrong, x is responsible for it.'

While these constructions still require detailed analysis, we acknowledge that the question-based approach does not naturally extend to them; as such, interrogative-related properties are not expected to arise. Nevertheless, this limitation does not undermine the viability of a question-based account for *wh*-conditionals. As illustrated in Figure 3, the question-based approach also incorporates operator − variable binding. In this respect, *wh*-conditionals may fall under the same broader umbrella as the constructions in (79) and (80), insofar as they all involve unselective binding. However, these constructions diverge in that they introduce bindees in non-uniform ways.

5. Conclusion and outlook

This paper draws on new evidence from Khorchin Mongolian to support the recently proposed question-based approach to *wh*-conditionals. Khorchin Mongolian features three types of *wh*-constructions—*wh*-questions, *wh*-indefinites, and *wh*-conditionals. Morphologically, *wh*-conditionals are more closely aligned with *wh*-questions than with *wh*-indefinites, as the *wh*-expressions in the first two constructions cannot co-occur with the indefinite determiner required for *wh*-indefinites. This pattern is consistent with the predictions of the question-based approach, which posits that a *wh*-conditional is formed by linking two *wh*-questions.

The present study further demonstrates that not all *wh*-constructions share the same underlying source. Some non-interrogative *wh*-constructions are in fact derived from interrogative ones, while others originate independently. This distinction aligns more closely with the cross-linguistic distribution of non-interrogative *wh*-constructions. In particular, for a language where all *wh*-expressions are inherently interrogative, non-interrogative *wh*-constructions can still emerge through transformations of *wh*-questions. English exemplifies this pattern: although *wh*-expressions in English are generally considered interrogative in nature, the language exhibits two non-interrogative *wh*-constructions—*wh*-free relatives and *wh*-unconditionals, as illustrated in (81) and (82).

(81) Joshua ate what Mary cooked. *wh*-free relatives

(82) No matter which route we take, we'll get to the beach eventually. *wh*-unconditionals

Although these two constructions are interpreted non-interrogatively, they share key properties with *wh*-questions. As a result, they have been argued to be derived from *wh*-questions (Rawlins 2013; Chierchia & Caponigro 2013; Xiang 2020). Typologically, therefore, *wh*-conditionals should be grouped with *wh*-free relatives and *wh*-unconditionals rather than with *wh*-indefinites. However, this prediction in typology also raises another question: why are *wh*-free relatives absent in Mandarin and Mongolian, and why are *wh*-conditionals not permitted in English? Addressing this issue requires a more comprehensive cross-linguistic comparison of these *wh*-constructions.

A A dynamic alternative analysis

In this appendix, we outline a specific implementation of the non-clausal answerhood operator, drawing primarily from Li (2021). As this topic is not central to the main argument of the paper, we refer interested readers to that work for a more detailed formal treatment.

According to Li (2021), the canonical Hamblin approach requires an upgrade to a dynamic framework to derive the meaning of *wh*-conditionals from interrogative semantics. The simplest way to implement a dynamic version of the Hamblin approach is to propose that a *wh*-expression evokes a set of *dynamic* individuals, and consequently, a *wh*-question denotes a set of *dynamic* propositions. In any dynamic framework, a dynamic proposition is treated as a context-change potential, formally modeled as a function from an information state (info-state, henceforth) to a set of info-states. Within this dynamic framework, the meaning of the *wh*-question in (69) can be redefined as shown in (83). While the derivation can also be carried out using η and $\circledast$, we omit the derivational details here to avoid unnecessary complexity. Readers interested in the specifics can refer to Li (2021).

(83) $[\![\text{Bat}^{v}\ \text{yuu}^{u}\ \text{id-sen be}]\!]^{w_0}_{d}$

$$= \left\{ \lambda w \lambda g. \{ g^{u \to x}_{v \to \mathsf{b}} \mid \mathsf{ate}_w(x)(\mathsf{b}) \} \;\middle|\; x \in \mathsf{thing}_{w_0} \right\} \qquad \text{Type: } \mathsf{s}\mathbf{t} \to \mathsf{t}$$

Info-states are modeled as assignments pf type g (e.g., g, h, g', h', etc.), and hence a dynamic proposition is of type $\mathsf{g} \to \mathsf{g} \to \mathsf{t}$, abbreviated as $\mathbf{t}$. Specifically, a dynamic proposition comprises a truth condition and the new variables (e.g., u, v, u', v', etc.) it adds to the input assignment. These variables are known as 'discourse referents' introduced by proper names, indefinites, *wh*-expressions, and definite descriptions. Accordingly, in (83), *Bat* and *yuu* introduce the discourse referents v and u respectively.

Discourse referents enable their introducers to remain accessible even after clausal meaning has been composed. This leads to a specific function ϵ_u known as 'Existential Disclosure', which maps a dynamic proposition ϕ to a dynamic predicate, as defined in (84) (Dekker 1993; Chierchia 1992, 2000). In essence, the function ϵ_u is anaphoric to a discourse referent u and retrieve its values.

(84) $\epsilon_u := \lambda\phi\lambda x\lambda w\lambda g.\{h \in \phi(w)(g) \mid h_u = x\}$ Type: $\mathbf{t} \to \mathsf{e} \to \mathbf{st}$

Moreover, following Dayal (1996), an interrogative meaning can be mapped to a propositional concept using an answerhood operator, defined in (85), which identifies a function from a world w to the weak exhaustive answer in w.

(85) $\mathbf{A}(Q) := \lambda w.\phi$ s.t. $\phi \in Q$, ϕ is true, and maximally informative in w.

Composing ϵ with $\mathbf{A}$ gives rise to a function mapping an interrogative meaning to a dynamic predicate, as shown in (86). In this case, ϵ_u must be co-indexed with a *wh*-expression. For concreteness, applying this function to (83) yields (87), which essentially characterizes the food that Bat ate in a world w.

(86) $\epsilon_u \circ \mathbf{A} = \lambda Q\lambda x\lambda w.\epsilon_u(\mathbf{A}(Q)(w))(x)(w)$ Type: $(\mathbf{st} \to \mathsf{t}) \to \mathsf{e} \to \mathbf{st}$

(87) $\epsilon_u \circ \mathbf{A}\ [\![\text{Bat}^v \text{ yuu}^u \text{ id-sen be}]\!]^{w_0}_d$

$= \lambda x\lambda w\lambda g.\{h \in \mathbf{A}[\![\text{Bat}^v \text{ yuu}^u \text{ id-sen be}]\!](w)(w)(g) \mid h_u = x\}$

$= \lambda x\lambda w\lambda g.\{g^{u\to x}_{v\to \mathsf{b}} \mid \mathsf{eat}_w(x)(\mathsf{b}) = \mathbf{A}_w[\![\text{Bat}^v \text{ yuu}^u \text{ id-sen be}]\!](w)\}$

Intuitively, the resultant dynamic predicate characterizes the so called 'short answer' to the *wh*-question. For example, if Bat ate beef in w_0 and lamb in w_1, the short answer to the question would be *beef* in w_0 and *lamb* in w_1, respectively. Based on (87), the pairs (w_0, b) and (w_1, l) lead to a dynamically true proposition. In this sense, $\epsilon_u \circ \mathbf{A}$ can be uderstood as another answerhood operator generating the short answer to a question. Accordingly, we represent $\epsilon_u \circ \mathbf{A}$ as $\mathbf{SA}$.

Returning to *wh*-conditionals, the *wh*-clauses involved retain their interrogative meanings, denoting sets of dynamic propositions. When embedded within a conditional, the two *wh*-clauses are shifted to predicate meanings through the operator $\mathbf{SA}$. These dynamic predicates serve as arguments for the modal/adverbial operator $\mathfrak{M}$, introduced by the conditional construction and defined in (75).

(88) Bat yuu id-sen bol, Dorž yuu id-sen.
Bat.NOM what eat-PST COND Dorž.NOM what eat-PST
'For everything x, if Bat ate x, Dorž ate x.'

(89) $[\![\mathbb{NEC}]\!]_{x,w}[\mathbf{SA}_u[\![\text{Bat}^v \text{ yuu}^u \text{ ate}]\!](x)(w)] : \mathbf{SA}_{u'}[\![\text{Dorž}^{v'} \text{ yuu}^{u'} \text{ ate}]\!](x)(w)$

In prose: For every possible world w and individual x such that x is the short answer to the question *what did Bat eat* in w, x is also the short answer to the question *what did Dorž eat* in w.

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Lists of abbreviations

NOM	nominative case marker
ACC	accusative case marker
DAT	dative case marker
GEN	genitive case marker
INS	instrumental case marker
PST	past tense marker
NPST	non-past tense marker
FUT	future marker
INF	infinitive marker
NEG	negative marker
SG	singularity marker
COMP	complementizer
Q	question particle
PQ	polar question particle
PFV	perspective marker
PASS	passive marker
CL	classifiers
COND	conditional marker
COV	coverb
Nec	Covert Necessity Modal
A	Answerhood Operator
SA	Short Answerhood Operator

Editors' note

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